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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2008 question paper

0625 PHYSICS

0625/31 Paper 31 (Extended Theory), maximum raw mark 80

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2008 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.

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NOTES ABOUT MARK SCHEME SYMBOLS AND OTHER MATTERS

B marks are independent marks, which do not depend on any other marks. For a B mark to be scored, the point to which it refers must actually be seen in the candidate's answer.

M marks are method marks upon which accuracy marks (A marks) later depend. For an M mark to be scored, the point to which it refers **must** be seen in a candidate's answer. If a candidate fails to score a particular M mark, then none of the dependent A marks can be scored.

C marks are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, provided subsequent working gives evidence that they must have known it. e.g. if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows he knew the equation, then the C mark is scored.

A marks are accuracy or answer marks which either depend on an M mark, or which are one of the ways which allow a C mark to be scored.

c.a.o. means "correct answer only".

e.c.f. means "error carried forward". This indicates that if a candidate has made an earlier mistake and has carried his incorrect value forward to subsequent stages of working, he may be given marks indicated by e.c.f. provided his subsequent working is correct, bearing in mind his earlier mistake. This prevents a candidate being penalised more than once for a particular mistake, but **only** applies to marks annotated "e.c.f."

e.e.o.o. means "each error or omission".

brackets ( ) around words or units in the mark scheme are intended to indicate wording used to clarify the mark scheme, but the marks do not depend on seeing the words or units in brackets e.g. 10 (J) means that the mark is scored for 10, regardless of the unit given.

underlining indicates that this must be seen in the answer offered, or something very similar.

OR/or indicates alternative answers, any one of which is satisfactory for scoring the marks.

Spelling Be generous about spelling and use of English. If an answer can be understood to mean what we want, give credit.

Significant figures Answers are acceptable to any number of significant figures $\geqslant 2$, except if specified otherwise, or if only 1 sig. fig. is appropriate.

Units It is expected that all final answers will have correct units. Deduct one unit penalty for each incorrect or missing unit, maximum 1 per question. No unit penalty if unit is missing from final answer but is shown correctly in the working.

Fractions These are only acceptable where specified.

Extras Ignore extras in answers if they are irrelevant; if they contradict an otherwise correct response or are forbidden by mark scheme, use right + wrong = 0

Ignore Indicates that something which is not correct is disregarded and does not cause a right plus wrong penalty.

Not/NOT Indicates that an incorrect answer is not to be disregarded, but cancels another otherwise correct alternative offered by the candidate i.e. right plus wrong penalty applies.

Work which has been crossed out, but not replaced, should be marked as if it had not been crossed out.

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1	**(a)**	**(i)**	any mention of force or weight ignore mass	C1	
			Force to left > force to right)		
			OR resultant force) any 1	A1	
			OR unbalanced force)		
			OR weight > friction)		
		(ii)	to overcome/compensate for friction/resistance	B1	
	(b)		2/2.5 or 4/5 etc. or F/a or F = ma	C1	
			0.8 kg	A1	
	(c)		0.7/0.8 e.c.f. from **(b)**	B1	
			0.875 (m/s^2) e.c.f. from **(b)** could be scored on table (no unit needed)	B1	
	(d)	**(i)**	v = at or 0.5×1.2	C1	
			0.6 m/s	A1	
		(ii)	any velocity × time or speed × time	C1	
			0.36 m c.a.o. (note: 0.72 m gets C1, A0)	A1	[11]
2	**(a)**		two masses chosen with ratio 2:1 or 3:1 or 3:2	M1	
			chosen masses in correct holes to balance	A1	
	(b)		disc does not rotate/is balanced/in equilibrium/no movement	B1	
			NOT spin the disc NOT anything to do with calculating moments		
			NOT when disturbed, returns to original position		
	(c)		moment of one mass correct (ignore units)		
			accept mass × distance calculated	B1	
			equal answers	B1	
	(d)		correct addition of masses/weights, including 200g	B1	
			any mass correctly converted to N	B1	[7]
3	**(a)**	**(i)**	hdg or $70 \times 1050 \times 10$	C1	
			735 000 Pa or 7.35×10^5 Pa accept N/m^2 for Pa	A1	
		(ii)	8.35×10^5 Pa OR his **(a)(i)** + 1.0×10^5 accept N/m^2 for Pa	B1	
	(b)		pressure × area or P = F/A or $6.5 \times 10^5 \times 2.5$	C1	
			1.625×10^6 N	A1	
	(c)		because density is less accept new calculation of pressure		
			OR because salt water is denser	B1	[6]

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4 **(a)** typical random path drawn, at least 3 abrupt changes of direction B1

(b) air molecules hit dust particles in all directions/move it in all directions B1
just as likely to be up as down B1
(allow marks scored on diagram)

(c) random movements smaller OR slower movement
OR less energy OR movement decreases B1 [4]

5 **(a)** **(i)** funnel no longer giving heat to ice OR ice at M.P./constant temp
OR heater reached max temp B1

(ii) inside of large pieces could be well below freezing point)
OR smaller air gaps if pieces smaller) any 1 B1
OR better contact between heater and ice)
OR to ensure heat from heater only goes to the ice)
OR larger surface area)
Ignore ice melts faster

(b) mass of beaker NOT mass of ice NOT mass of water B1
mass of beaker + water B1
(apply ✓ + × = 0 for extras other than power & time)

(c) (mass of ice melted by heater = 16.3 – 2.1) = 14.2 g C1
ml in any form, words, symbols or numbers C1
Wt or Pt in any form, words, symbols or numbers accept VIt C1
338 J/g OR 338 000 J/kg c.a.o A1 [8]

6 **(a)** light of one colour/frequency/wavelength B1

(b) n = sin$r$/sin$i$ OR n = sin$i$/sin$r$ in any form C1
sin$r$/sin30 = 1.49 OR sin$r$ = 1.49 × sin30 C1
48.0° – 48.2° A1

(c) ray at angle >30° and <60° to normal, by eye, correct way **NO** e.c.f. B1
Ignore any angles or labelling

(d) colours/spectrum would appear OR range of angles (ignore "rainbow")
OR dispersion OR ray splits up B1

(e) 90° approx (accept any value 80° to 90°) B1

(f) (totally internally) reflected OR T.I.R. ignore not refracted B1 [8]

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7	**(a)**	clear attempt at arcs of circles, at least 3		B1	
		same wavelength as incoming waves, by eye (ignore shape ignore distance to first wave)		B1	
		centre of curvature of arcs at centre of gap, by eye		B1	
	(b)	speed/wavelength or 20/2.5 or $v = f\lambda$		C1	
		8 Hz or 8 s^{-1} or 8 waves/second		A1	
	(c)	his **(b)** or "the same"		B1	[6]
8	**(a)**	changes a.c. to d.c. OR rectifies a/c OR allows current to flow one way only OR prevents current flowing backward		B1	
	(b)	$I \times t$ or 2×12 or $2 \times 12 \times 60 \times 60$ or amps × seconds		C1	
		24 Ah or 86 400 C or 86 000 C		A1	
	(c)	emf = J/C OR energy converted/work done per unit charge/coulomb OR W/A OR volts/p.d. when no current in circuit		C1	
		12 J of energy are delivered/needed for every coulomb of charge OR 12 W is the power to drive a current of 1 A		A1	
	(d)	**(i)** series connection shown, any recognisable symbols		B1	
		(ii) total power = 16 W OR 8/6		C1	
		1.33 A accept fraction c.a.o.		A1	
		(iii) any power × any time or $16 \times 60 \times 60$ or IVt or $8 \times 60 \times 60$		C1	
		57 600 J or 0.016 kWh or 28 800 J or 0.008 kWh		A1	[10]
9	**(a)**	pump water to higher level storage or heat water or charge accumulators/batteries ignore charge capacitor NOT generator	) any one	B1	
	(b)	less/no energy/power/heat loss OR to reduce current OR to allow thinner cables OR more efficient NOTHING ELSE		B1	
	(c)	I^2R		B1	
	(d)	$N_s/1200 = 32000/1100$ OR $N_1/N_2 = V_1/V_2$ in any arrangement		C1	
		34 880 or 34 900 or 34 909 or 34 910 or 35 000		A1	
	(e)	input power = output power or $V_1I_1 = V_2I_2$		C1	
		current = power/voltage in any form, words, symbols or numbers		C1	
		25 A		A1	[8]

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10 (a) (i) LDR correctly identified B1

(ii) lamp correctly identified B1

(iii) transistor correctly identified B1

(b) (ignore anything that is in terms of currents)
resistance of LDR becomes high M1
LDR gets larger share of the voltage OR voltage across LDR gets bigger A1
transistor switches/turns lamp on A1 [6]

11 (a) A cathode OR electron gun
B Y plates OR vertical deflection plates
C X plates OR horizontal deflection plates
D screen OR fluorescent/phosphor OR tube NOT glass
4 correct B2, 3 or 2 correct B1 B2

(b) A; idea of releasing electrons/thermionic emission B1
B; move the electron beam vertically B1

(c) (i) y-plates/y-input or B **NO** e.c.f. B1

(ii) x-plates/x-input or C **NO** e.c.f. B1 [6]